

JCC Dev

Web Application. Mobile

1. User Roles

Role	Scope	Key Permissions
Managing Director	All office locations	Cross-location aggregated dashboard. No task creation or edit rights.
Senior QS (Admin)	Own office / project	Create & assign tasks, review submissions, approve BOQ items, view all office dashboards.
Quantity Surveyor (User)	Assigned tasks only	View & execute tasks, log measurements, mark attendance, raise queries.
System	All	Auto-notifications, status transitions, rate calculations, escalation triggers.

1. Core Modules

Module	Purpose	Key Output	
Task Management	Taking-off assignment and tracking	Completed measurement packages, BOQ-ready quantities	2
Attendance Tracking	Daily QS availability	Attendance records, risk flags on tasks, leave logs	1
Rate Benchmarking	Planning vs actual productivity	Efficiency %, historical rate library, proposal support data	Not important right now

1. Detailed Workflow

MODULE A — Task Creation & Assignment (Senior QS) (PM in IT)

The Senior QS creates a structured task package for each taking-off scope:

- Create Project — name, contract reference, office location
- Break scope into Tasks — e.g. 'Concrete volume for footing F1'
- For each task, set: assigned QS, deadline, priority (Critical / High / Standard)

- Set planned productivity rate — e.g. 50 m² / day, 30 m³ / day (set by the SQS) (shouldn't be visible to junior people in the future)

	Task ID format: TASK-[OfficeCode]-[ProjectCode]-[Trade]-[YYYYMMDD]-[Seq] Example: TASK-CMB-BLKC-CONC-20260424-003
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MODULE B — Task Execution (QS Surveyor) (SEs)

The assigned QS Surveyor executes the taking-off and submits measurements:

- Open assigned task — status moves to IN PROGRESS
- Update daily work done as the work finishes (date and time)
- *Raise a Query Log entry for any drawing discrepancy or scope ambiguity (not important)*

Trigger	Status Change
QS acknowledges and opens task	ASSIGNED → IN PROGRESS
<i>QS raises a query</i>	<i>IN PROGRESS + QUERY RAISED flag (parallel)</i>
<i>SQS responds to query</i>	<i>QUERY RAISED flag cleared → IN PROGRESS</i>
<i>QS requests deadline extension</i>	<i>IN PROGRESS → EXTENSION REQUESTED</i>

MODULE C — Attendance Tracking (QS Surveyor daily)

Each QS marks their daily attendance status at the start of the working day:

Status	System Action
Present	No action — tasks proceed normally
Absent	All assigned active tasks flagged as AT RISK; <i>Senior QS notified immediately</i>
Half-day	Tasks with same-day deadlines flagged; <i>SQS notified</i>
On Leave (with reason)	Tasks flagged; <i>SQS notified; optional auto-delay or reassignment triggered</i>

When a QS is marked absent or on leave, the SQS receives a notification listing all affected tasks with their current status and deadline. The SQS can then:

- *Manually reassign affected tasks to another QS*

- Approve an automatic deadline extension
- Place affected tasks in ON HOLD status pending the QS return

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MODULE D — Planning vs Actual Productivity Tracking

For every submitted task the system automatically calculates:

Metric	Formula / Source
Planned Rate	Set by SQS at task creation (e.g. 50 m ² /day)
Actual Rate	Quantity completed ÷ Total time logged (from mandatory time log)
Efficiency %	Actual Rate ÷ Planned Rate × 100
Variance	Actual Rate – Planned Rate (with unit)

Metrics are computed at three levels: per task, per QS Surveyor (rolling average), and per activity type (trade element). This builds a firm-specific productivity library over time.

1. Review & Approval (Senior QS)

Triggered when a task reaches SUBMITTED status

Review Outcome	Status Set	Next Action
No changes required	APPROVED	Proceed to finalisation and BOQ export
Minor corrections needed	REVISION REQUIRED	Return to QS with marked comments; QS resubmits
Major rework needed	REJECTED	Task reassigned or escalated; new target date issued
Waiting on third-party info	ON HOLD	Status frozen; SQS resolves dependency; SLA paused

1. Dashboards

Managing Director — Multi-Location Overview

Read-only aggregated view across all office locations. No task-level detail.

Panel	Content
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Location Selector	All Offices / individual office filter (e.g. Colombo, Kandy, Galle)
Project Pipeline per Location	Active projects, task completion %, delayed task count, tasks at risk
QS Headcount & Availability	Current week attendance summary per office
Monthly Completion Rate	Tasks closed vs tasks created — trend chart per office

Senior QS — Office Dashboard

- *Task status breakdown — pending, in progress, submitted, delayed, at risk*
- *QS availability today — attendance summary for the team*
- *Open queries requiring SQS response*
- *Team workload — tasks per QS (current + pipeline)*
- *Productivity comparison — planned vs actual rates by activity type*
- *Revision rate by QS Surveyor — quality indicator*
- *Bottlenecks — tasks blocked by ON HOLD or EXTENSION REQUESTED*

QS Surveyor — Personal Dashboard

- *My assigned tasks — sorted by priority and due date*
- *Upcoming deadlines — 7-day and 14-day view*
- *Personal productivity stats — efficiency % vs firm average*
- *Completed tasks — historical archive with approved quantities*

1. Key Data Structure

Entity	Key Fields
Office / Location	Office ID, name, city, country, managing SQS
Project	Project ID, name, contract ref, office, status
Task	Task ID, project, trade, drawings, BOQ range, assignee, SQS, planned rate, status, revision count
User (QS / SQS / MD)	User ID, name, role, office, active flag
Attendance Record	User ID, date, status (Present/Absent/Half/Leave), reason

Measurement Output	Task ID, BOQ item code, quantity, unit, dimension sheet ref
Time Log	Task ID, user ID, date, hours logged — mandatory on submission
Productivity Rate	Task ID, activity type, planned rate, actual rate, efficiency %, date
Query Log	Task ID, drawing ref, description, impact, raised by, response, resolved flag
Audit Trail	Task ID, user, timestamp, action, comment — append-only

1. **Notification & Escalation Rules**

Trigger	Recipient	Channel
<i>Task assigned</i>	<i>QS Surveyor</i>	<i>In-app + Email</i>
<i>QS marked absent / on leave</i>	<i>Senior QS + affected task flag</i>	<i>Email immediately</i>
<i>3 days before deadline (no submission)</i>	<i>QS Surveyor + SQS</i>	<i>Email reminder</i>
<i>Deadline passed (no submission)</i>	<i>QS Surveyor + SQS + MD dashboard flag</i>	<i>Email escalation</i>
<i>Query raised</i>	<i>Senior QS</i>	<i>In-app + Email</i>
<i>Query response given</i>	<i>QS Surveyor</i>	<i>In-app + Email</i>
<i>Task returned for revision</i>	<i>QS Surveyor</i>	<i>In-app + Email</i>
<i>Task approved</i>	<i>QS Surveyor + Coordinator</i>	<i>In-app notification</i>
<i>Extension request raised</i>	<i>Senior QS</i>	<i>In-app + Email for decision</i>